

CLOUD COMPUTING CONCEPTS

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Part 2 INTRODUCTION

GOAL OF C₃ - PART 2

- Course about the internals of cloud computing
- **Distributed systems and algorithms** that underlie today's cloud computing technologies
- We'll discuss
 - Concepts
 - Techniques
 - Industry systems, including open-source (from the inside)
- Builds on “Cloud Computing Concepts (C3) – Part 1”
 - C3 – Part 1 is a prerequisite for C3 – Part 2
 - We'll be reusing and relying on many concepts you learnt in Part 1 throughout Part 2

SYLLABUS FOR C₃ – PART 2

- Classical algorithms: Leader Election, Mutual Exclusion, Scheduling
- Scalability: Concurrency control, Replication Control
- Trending Areas: Stream processing, Graph processing, Structure of Networks, Sensor Networks
- Miscellaneous: Distributed File systems, Distributed shared memory, Security, Datacenter outage studies
- Fun: Interviews with leading managers and researchers, from both industry and academia

WHAT WE'LL BE RELYING ON FROM C₃ – PART 1

- Introduction: Clouds, Mapreduce, Key-value stores
- Classical Precursors: Peer-to-peer systems, Grids
- Widely-used algorithms: Gossip, Membership, Paxos
- Classical algorithms: Time and Ordering, Snapshots, Multicast

EXERCISES

- 5 Homeworks
- 1 Programming Assignment (C++)
 - Implement a fault-tolerant key-value store inside an emulator
 - Builds on Programming Assignment from C3 – Part 1
- 1 Exam

ONWARD!

- Cloud computing is an exciting area to be studying, very dynamic and continuously changing
- Come, let's complete our journey through the landscape.